

Precalc Lesson 1.9 Homework

Answer Key

1. 33 school buses can transport 650 students.
2. $g(x) = -3(x - 11) = -3x + 33$
3. 8
4. f^{-1} gives the number of years you must work at a company, t , in order to earn m dollars. The amount of money you want to earn is the input and the years you must work is the output.
5. C
6.
 - a. c
 - b. d
 - c. $d = \sqrt[3]{c - 5}$ or $d = (c - 5)^{\frac{1}{3}}$
7.
 - a. $t = \frac{q - 54}{-2.5}$
 - b. The input of f^{-1} is the volume of water in the sink in quarts, q , and the output of f^{-1} is the time, t .
 - c. 21.6 seconds
 - d. 7.2 seconds
8.
 - a. $g^{-1} = \frac{x^2 + 6}{2}, x \geq 0.$
 - b. $g(g^{-1}(x)) = \sqrt{2(\frac{x^2 + 6}{2}) - 6} = \sqrt{x^2} = x$
 $g^{-1}(g(x)) = \frac{(\sqrt{2x - 6})^2 + 6}{2} = \frac{2x - 6 + 6}{2} = x$